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**University of Colombo, Sri Lanka****University of Colombo School of Computing****Bachelor of Science in Computer Science**

First Year Examination — Semester II - 2021

SCS1214 — Operating Systems I

(2 Hours)

Answer All Questions

Number of Pages = 12

Number of Questions = 4

To be completed by the candidate**Index Number**

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Important Instructions

- The duration of the paper is 2 Hours.
- The medium of instructions and questions is English. Answer questions in English.
- This paper has 4 questions on 12 pages.
- Answer **all** questions.
- **Write your answers on and only on the space provided** on this question paper.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries and mobile phones, are **not allowed**.
- Non-programmable Calculators may be used.

To be completed by the examiners

1	
2	
3	
4	
Total	

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1. (a). The following code segment is written in the assembly instructions of the virtual machine used for the assignments of the SCS1214 course.

```
0 movv sp 90
1 movv a 1
2 movv b 2
3 call 18
4 add a b
5 out acc
6 halt
```

```
18 push a
19 push b
20 movv a 10
21 movv b 5
22 sub a b
23 mov a acc
24 out acc
25 jnz 22 a
26 pop b
27 pop a
28 ret
```

- i. What is the output when the above code is executed on the virtual machine?

[3 marks]

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- ii. Assume that the lines numbered 3 and 4 are interchanged as follows.

```
3 add a b
4 call 18
```

What is the output of this modified program?

[4 marks]

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iii. Assume that the lines numbered 26 and 27 are interchanged as follows.

26 pop a

27 pop b

Does this affect the output? Justify your answer.

[5 marks]

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(b). A process goes through the *new*, *ready*, *running*, *waiting*, and *terminated* states. The next state after a state transition of the process is given by the function $T(s, e)$, where s is the current state and e is an event.

i. If $T(T(T(x, e_1), e_2), e_3) = x$, give 3 possible states for x .

[3 marks]

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ii. There are 3 different events e_a , e_b , and e_c and 3 different states s_a , s_b , and s_c such that,

$$T(y, e_a) = s_a$$

$$T(y, e_b) = s_b$$

$$T(y, e_c) = s_c$$

What is the state y ?

[3 marks]

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(c). Following program is compiled and executed on an *x86* machine running Linux.

```
#include <stdio.h>
#include <sys/types.h>
#include <unistd.h>

int main()
{
    if(fork())
        if(fork())
            if(fork())
                printf("X\n");

    printf("Y\n");
    return 0;
}
```

i. How many X's are printed by this program?

[3 marks]

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ii. How many Y's are printed by this program?

[4 marks]

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2. (a). The structure of the producer process of the bounded-buffer problem is given below.

```
do{
    // Produce an item
    wait(X);
    wait(Y);
    // add the item to the buffer
    signal(Y);
    signal(Z);
}while (TRUE);
```

This solution uses a semaphore named `mutex` to ensure mutual exclusion and a semaphore named `full` initialized to 0 and the semaphore named `empty` initialized to the size of the buffer.

- i. What is X?

[3 marks]

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- ii. What is Y?

[3 marks]

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- iii. What is Z?

[3 marks]

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- iv. Write the code for the corresponding consumer process.

[5 marks]

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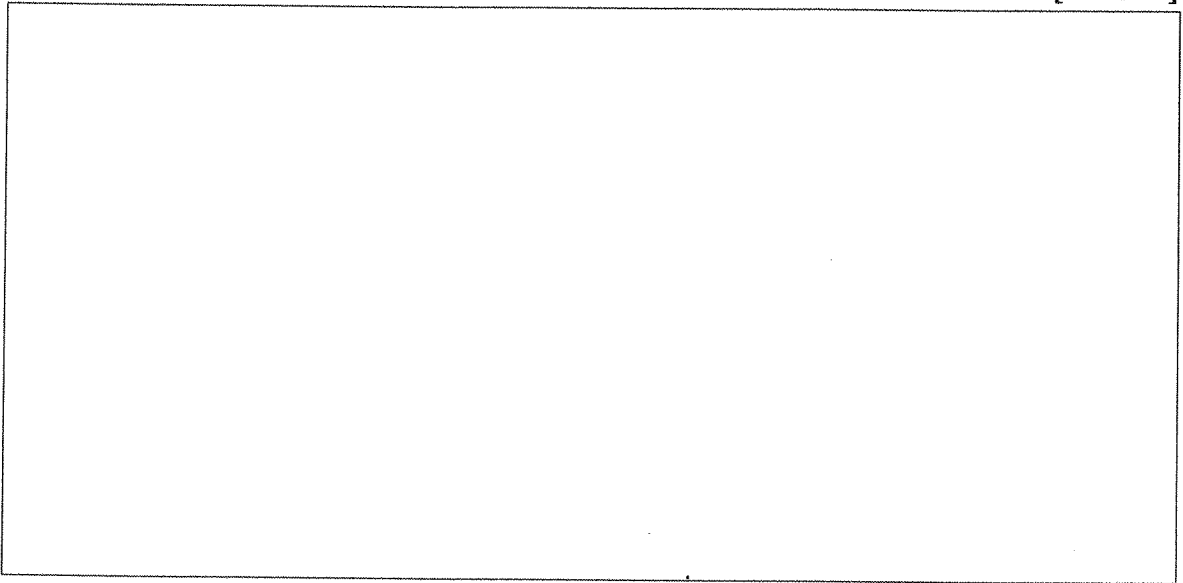
- (b). The source codes of the processes P1 and P2 are given bellow. S and Q are two semaphores initialized to 1.

```
// Process P1
do{
    wait(S);
    wait(Q);
    ...
    signal(S);
    signal(Q);
}while(true);
```

```
// Process P2
do{
    wait(Q);
    wait(S);
    ...
    signal(Q);
    signal(S);
} while(true);
```

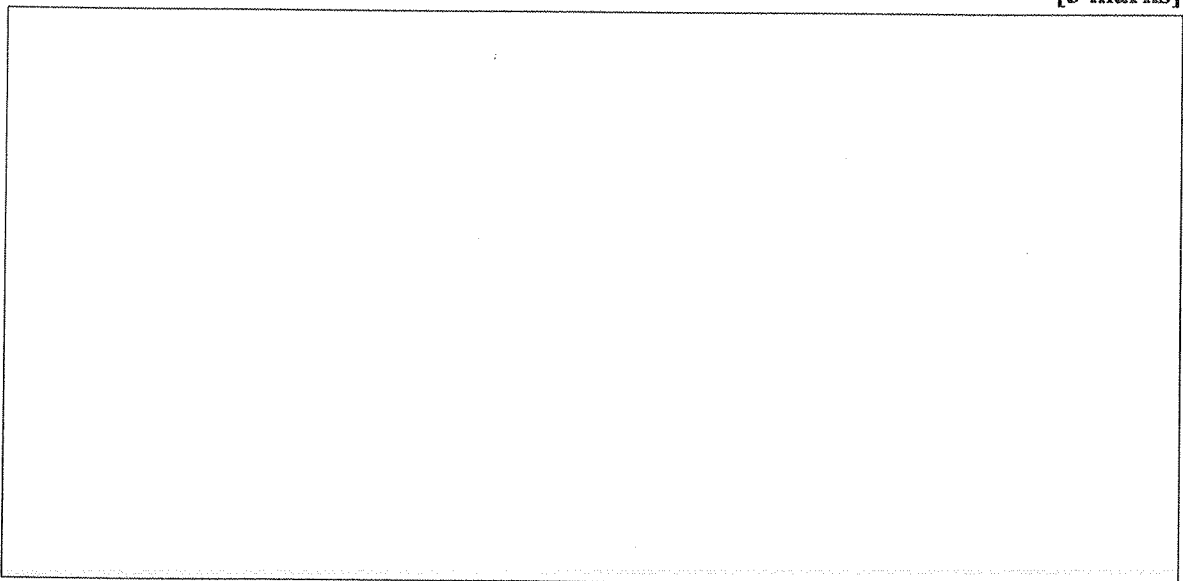
- i. Draw the resource allocation graph to depict a deadlock involving P1 and P2.

[5 marks]



- ii. Draw the wait-for graph to depict the above deadlock involving P1 and P2.

[3 marks]



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- | | Maximum Need | Current Allocation |
|-------|--------------|--------------------|
| P_0 | 10 | 6 |
| P_1 | 9 | 2 |
| P_2 | 4 | 3 |

[3 marks]

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3. (a). Explain the difference between preemptive and non-preemptive scheduling by providing examples for each category.

[5 marks]

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- (b). Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of burst time listed. In answering the questions, use non-preemptive scheduling, and base all decisions on the information you have at the time the decision must be made.

Process	P1	P2	P3	P4
Arrival time	0.0	0.5	1.0	1.3
CPU burst time	8	5	2	1

Note : Provide your calculation based answers in one decimal point without rounding off.

- i. What is the average turnaround time for these processes with the First-Come-First-Serve (FCFS) scheduling algorithm ?

[5 marks]

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- ii. What is the average turnaround time for these processes with the Shortest-Job-First (SJF) scheduling algorithm ?

[5 marks]

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- iii. What if the CPU is left idle for the first 1.5 time unit and then SJF scheduling algorithm is used. Compute the average turnaround time and waiting time. Please note that Processes are waiting during this idle time and their waiting time may increase.

[5x2 marks]

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4. (a). What is meant by Beladys anomaly ? What are the page replacement algorithms that will not encounter Beladys anomaly ?

[3 marks]

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- (b). Consider the following reference string from a address sequence after tracing a particular process.

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

- i. Compute the number of page faults for three (3) page frames for the following page replacement algorithms.

A. First-In-First-Out (FIFO) Algorithm

[3 marks]

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B. Optimal Algorithm

[3 marks]

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C. Least Recently used (LRU) Algorithm

[3 marks]

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D. What are two mechanisms that we can use to implement Least Recently used (LRU) on the system ?

[2 marks]

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ii. What is the difference between sequential and direct access methods ?

[2 marks]

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- iii. Assume you are given a system that consist of 32 bit addressing. Each data block is 1KB and pointer size is 4 byte in size. What is the maximum file size that can exist in the system?

Assume that there are 10 direct pointers to data blocks, 2 single indirect pointer, 1 double indirect pointer, and 1 triple indirect pointer defined in the file system.

[6 marks]

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- iv. What are three (3) use cases of virtualization software ?

[3 marks]

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